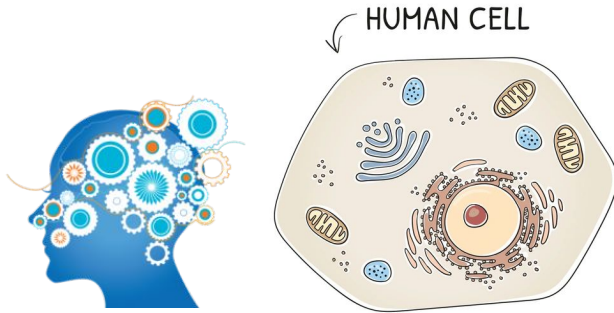


Gametes

Textbook pages: 247 - 252

Lesson objectives

- ❑ To define gametes as sex cells involved in sexual reproduction and recognise sperm and egg cells as male and female gametes.
- ❑ To explain why human gametes contain only 23 chromosomes and a zygote has 46.
- ❑ To describe alleles as different versions of the same gene.



How many chromosomes do most cells in the human body have?

Where did these chromosomes come from?

Keywords: gamete, sperm cell, egg cell, nucleus, cell membrane, cytoplasm, flagellum (tail), head, acrosome, mitochondria, jelly coat, fertilisation, zygote, allele, sex inheritance

Success Criteria		
Working towards	Working at	Working beyond
I can define gametes as sex cells involved in sexual reproduction and recognise sperm and egg cells as male and female gametes.	I can explain why human gametes contain only 23 chromosomes and a zygote has 46.	I can describe alleles as different versions of the same gene.

Gametes

Sex cells

What is the scientific term for 'sex cell'?

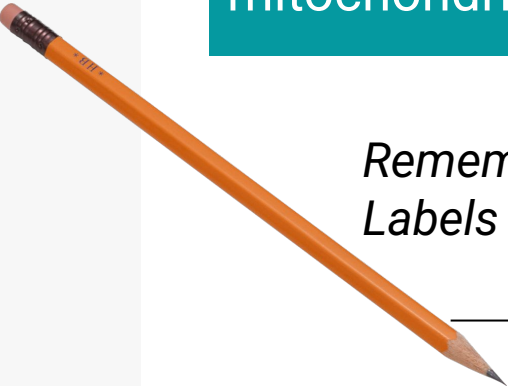
Draw and **label** human sex cells.

Extension: Describe and explain how egg and sperm cells are adapted to their function.

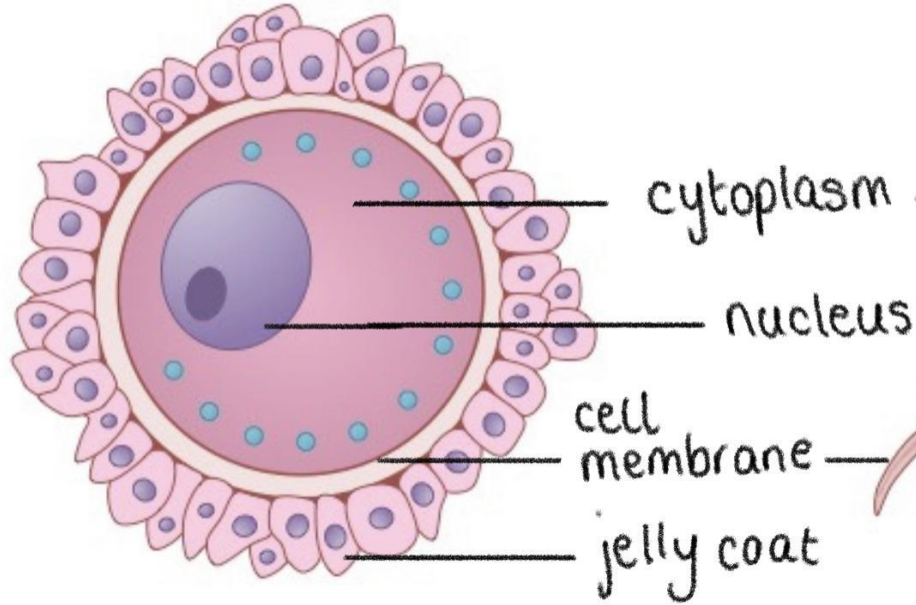
nucleus, cell membrane, cytoplasm, flagellum (tail), head, acrosome, mitochondria, jelly coat

*Remember that all drawings should be done with a pencil
Labels should be done with a straight line*

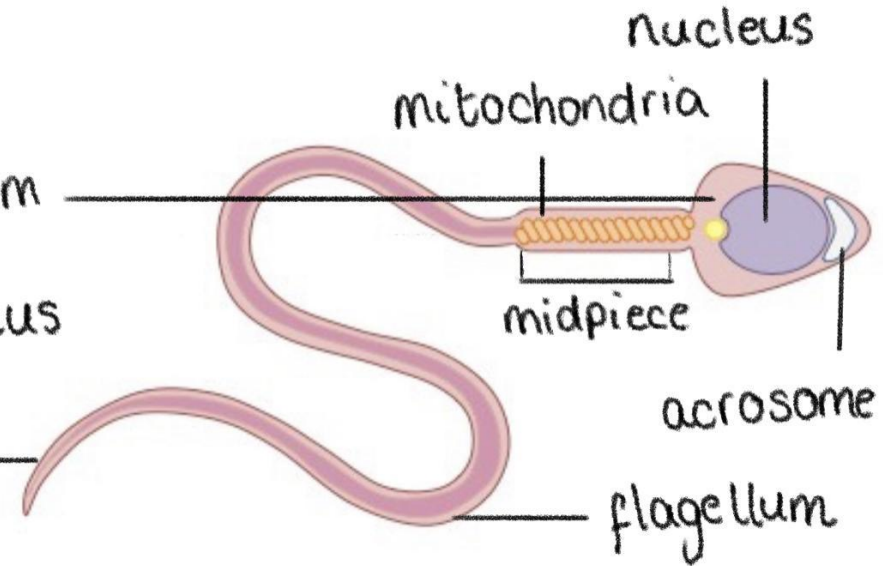
pencil



Human Egg (Ovum)



Human Sperm (Spermatozoa)



Sperm cells are actually 10 000 times smaller than egg cells!

Where are they produced?

GAMETE	ADAPTIVE FEATURE	REASON
SPERM	HAS A FLAGELLUM (TAIL)	ENABLES IT TO SWIM TO THE EGG
	CONTAINS ENZYMES IN THE HEAD REGION (ACROSOME)	TO DIGEST THROUGH THE JELLY COAT AND CELL MEMBRANE OF AN EGG CELL WHEN IT MEETS ONE
	CONTAINS MANY MITOCHONDRIA	PROVIDE ENERGY FROM RESPIRATION SO THAT THE FLAGELLUM CAN MOVE BACK AND FORTH FOR LOCOMOTION
EGG	CYTOPLASM CONTAINING A STORE OF ENERGY	PROVIDES ENERGY FOR THE DIVIDING ZYGOTE AFTER FERTILISATION
	JELLY LIKE COATING THAT CHANGES AFTER FERTILISATION	FORMS AN IMPENETRABLE BARRIER AFTER FERTILISATION TO PREVENT OTHER SPERM NUCLEI ENTERING THE EGG CELL

CONCEPT CHECK 1 – choose the correct statement



- A Male gamete is larger than female gamete
- B Male gamete is smaller than female gamete
- C Male gamete is the same size as the female gamete

CONCEPT CHECK 2 – choose the correct statement

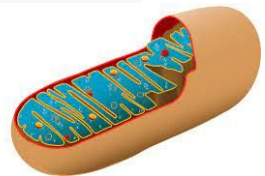


- A. Male gamete can swim while the female gamete cannot move/swim
- B. Female gamete can swim while the male gamete cannot move/swim
- C. Both gametes can swim
- D. Both gametes cannot move/swim

CONCEPT CHECK 3 – choose the correct statement



- A. Male gamete has lesser mitochondria and a tail
- B. Female gamete has a lots of mitochondria and a tail
- C. Male gamete has a lots of mitochondria and no tail
- D. Female gamete has lesser mitochondria and no tail

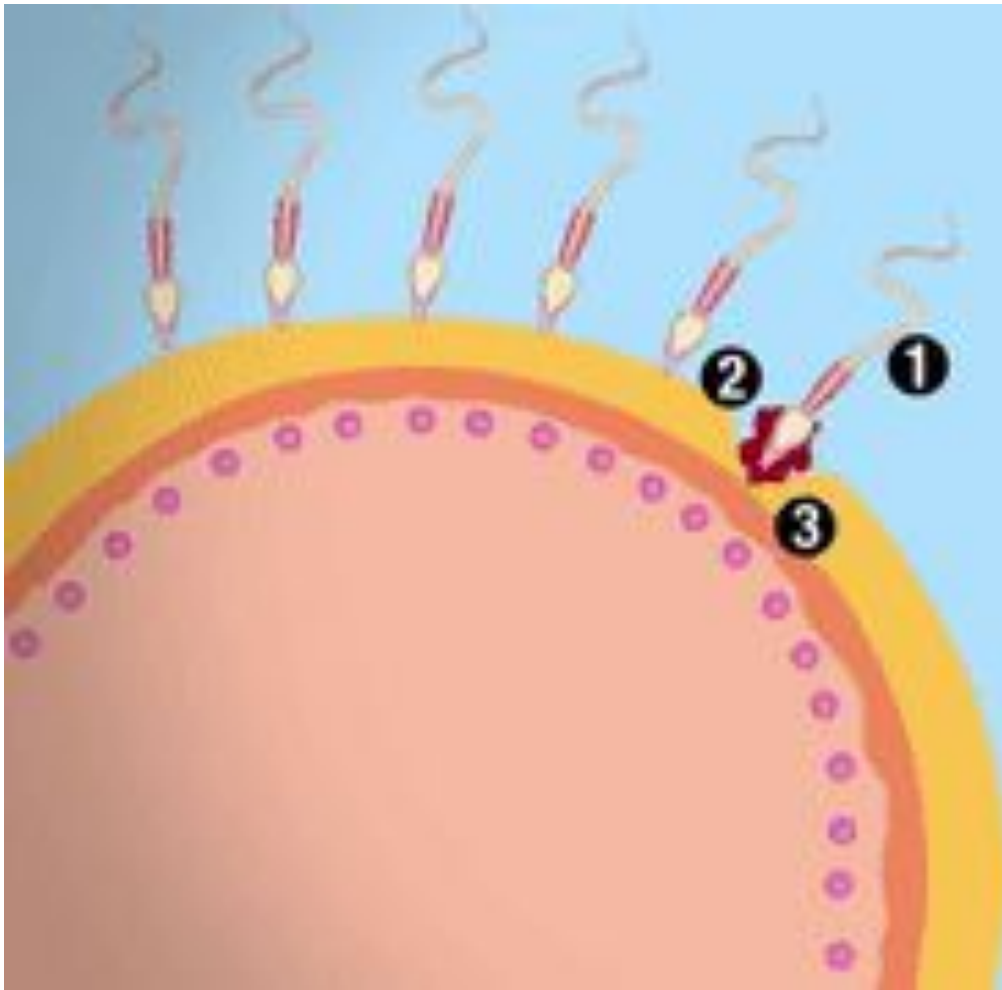


mitochondria

CONCEPT CHECK 4 – choose the correct statement



- A. Male gamete is produced in large numbers
- B. Female gamete is produced in small number
- C. Male gamete is produced in small number
- D. Female gamete is produced in large numbers

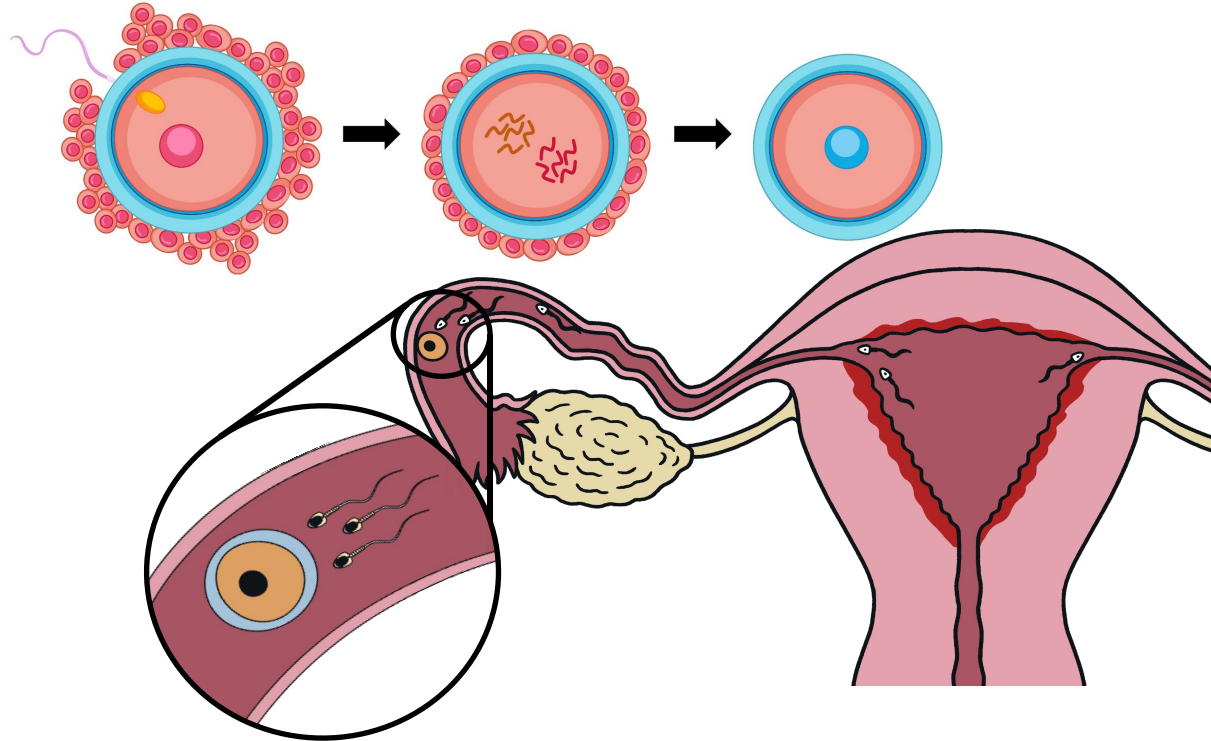


1. Sperm can reach the egg within 15-45 minutes of ejaculation, though some may take longer.
2. Enzymes released by the sperm help break down the jelly coat surrounding the egg.
3. The first sperm to break through the egg's outer layers and reach the egg's membrane releases its genetic material, initiating fertilisation.

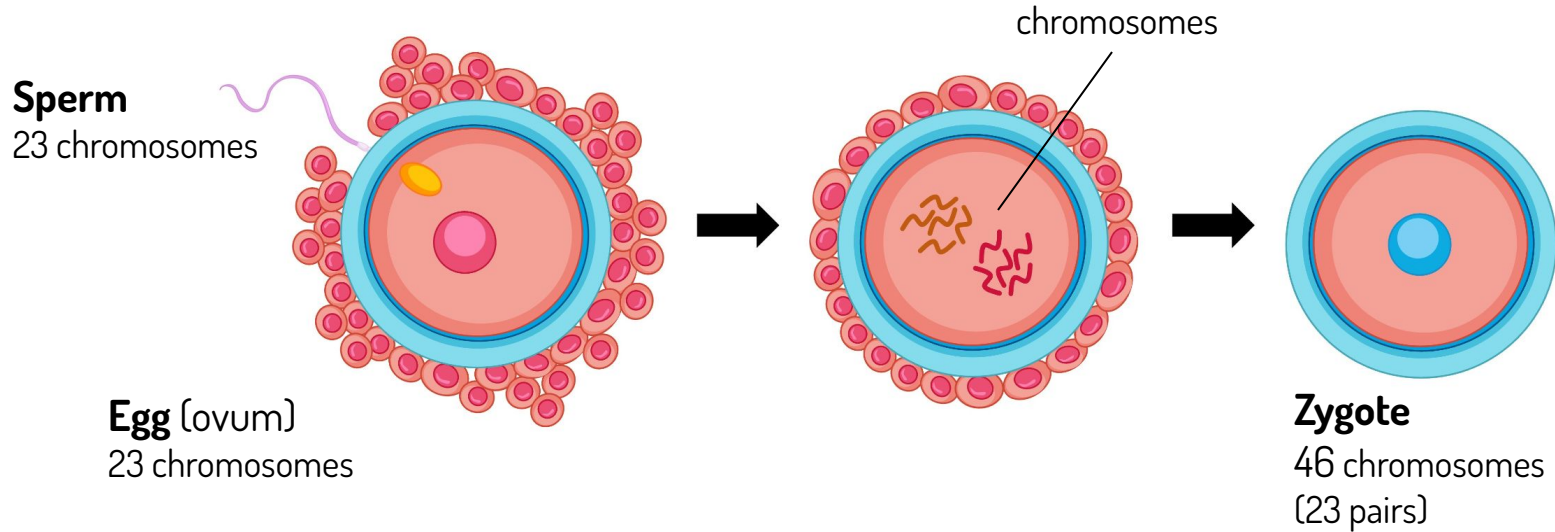
Fertilisation

Use the following keywords to construct a sentence that best describes fertilisation.

- ***Sperm cell***
- ***Fuse/ Fusion***
- ***Egg cell***
- ***Zygote***
- ***Nucleus***



Fertilisation

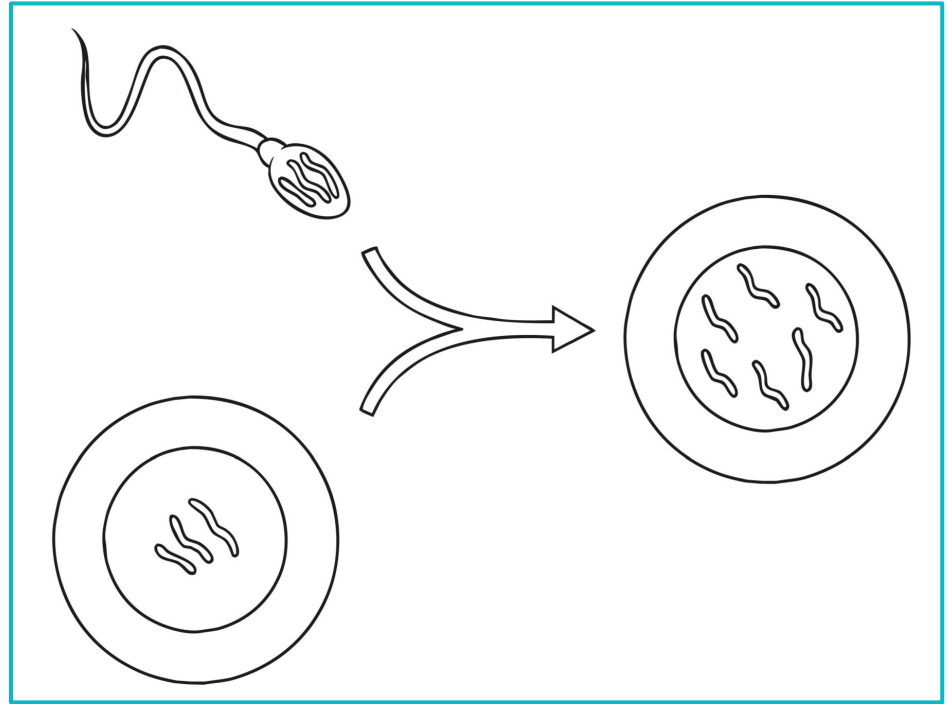


Fertilisation is the process that takes place when a **sperm cell nucleus** meets and **fuses** with an **egg cell nucleus** to produce a **zygote**.

Fertilisation

Copy this diagram in your books.

1. Add labels.
2. Using 2 colours, show how the new cell produced has 2 sets of chromosomes - one from each parent.



Fertilisation is the process that takes place when a **sperm cell nucleus** meets and **fuses** with an **egg cell nucleus** to produce a **zygote**.





1



2



3



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21



22



X



Y

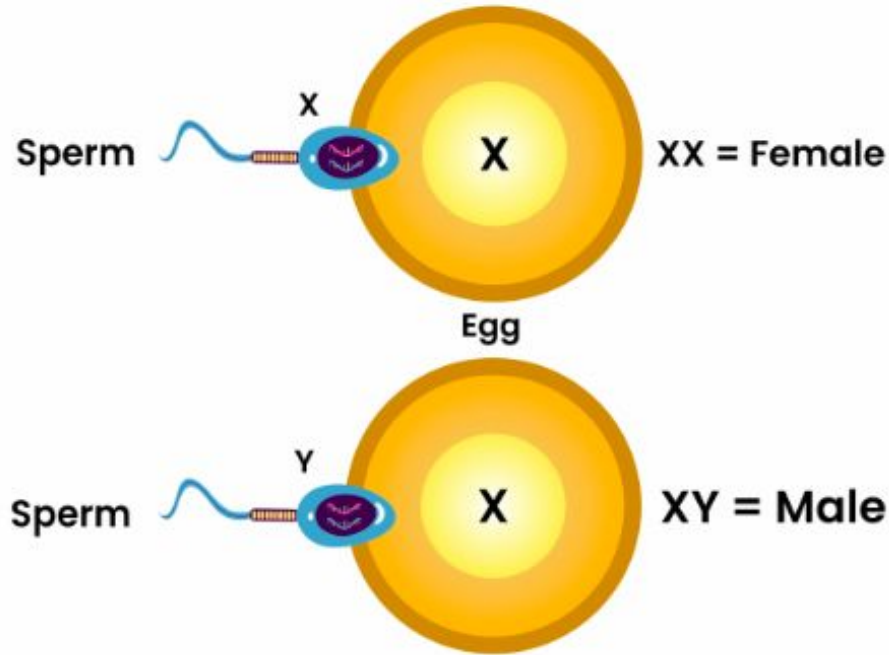
or



X X

Shown here are human chromosomes

Boy or girl?



Is it correct to say that the sex of a baby is determined by its father, not its mother?

Common misconception

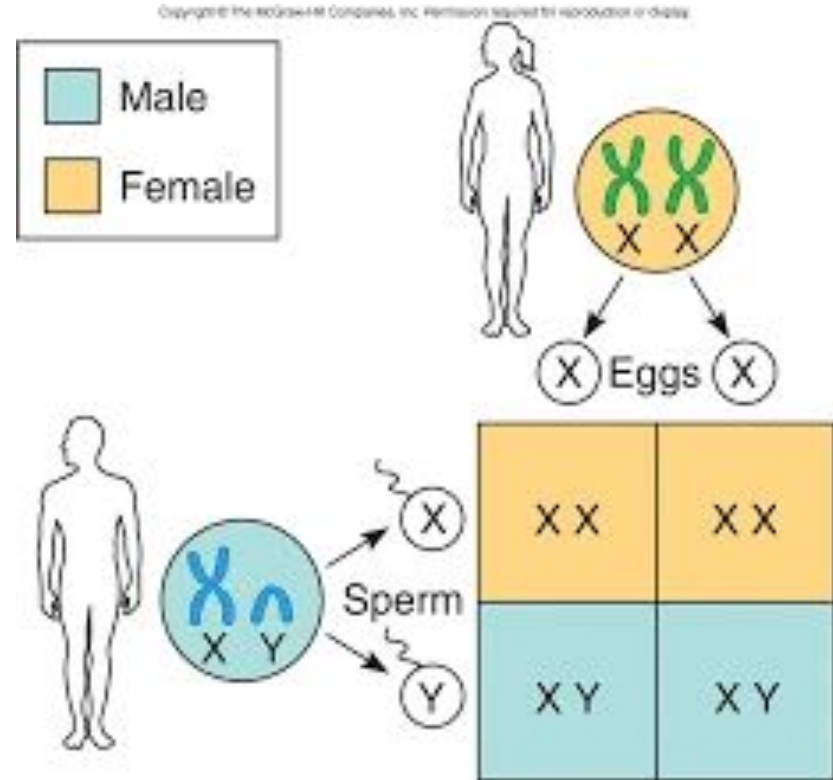
Some people think that sperm cells contain male chromosomes, and egg cells contain female chromosomes... this is **not** true!

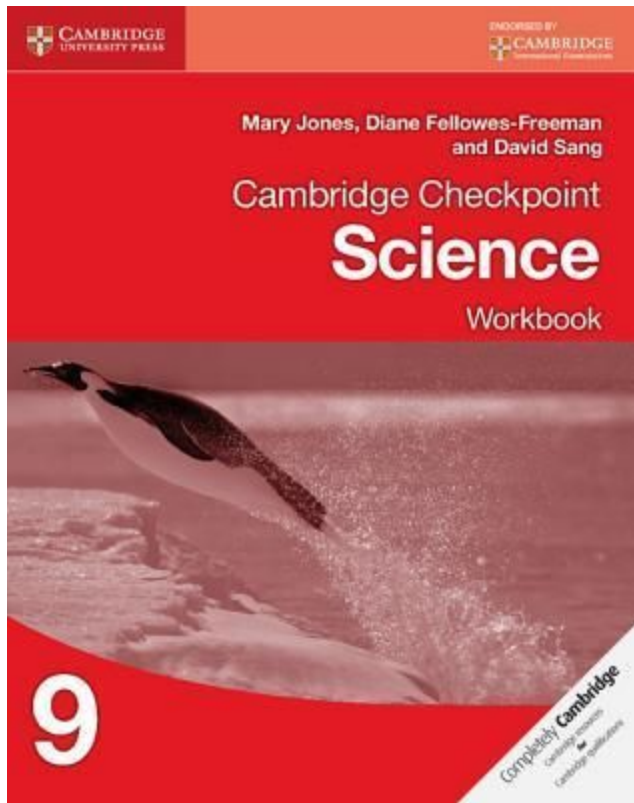
Inheritance

Inheritance means passing on DNA (as chromosomes containing genes) from parents to offspring.

Punnett squares can be used to calculate the probability of the genders of people.

We will learn more about these next lesson!





Complete:

Exercise **7.1** Chromosomes,
genes and DNA (pages 133+134)

Exercise **7.2** Egg cells and sperm
cells (pages 135 + 136)

Spot the mistakes



Most cells in a human body contain 23 chromosomes.	Egg cells have a protective jelly-like coating that contains acrosomes.	The nuclues of the sperm cell and the nuclues of the egg cell fusion together.
A person with one X and one Y chromosome is female.	Sperm cells can have either XY chromosomes or XX chromosome.	Usually, there are more girl babies born each year.

Spot the mistakes

Most cells in a human body contain 46 chromosomes.	Sperm cells' heads contain acrosomes with digestive enzymes.	The nucleus of the sperm cell and the nucleus of the egg cell fuse together.
A person with two X chromosomes is female.	Sperm cells can have either one X chromosome or one Y chromosome .	The numbers of boy and girl babies born each year is approximately equal .

